

1. LAYERED ARCHITECTURE

Layered Architecture organizes the system into horizontal layers such as Presentation, Business, and Data. It provides clear separation of concerns and is easy to understand. However, it can introduce performance overhead, tight coupling between layers, and difficulty in scaling individual components.

SUITABLE FOR COFFEE KIOSK

Suitable because the kiosk has clearly separated responsibilities such as touchscreen interaction, order processing, payment, receipt printing, and data storage.

2. MICROSERVICES ARCHITECTURE

Microservices Architecture divides the system into small, independently deployable services. Its advantages include independent scaling, fault isolation, technology diversity, and independent deployment. Its disadvantages include operational complexity, network latency, and data consistency challenges.

Suitability for Coffee Kiosk:

Possible, but it may introduce unnecessary complexity for the relatively focused functionality of the kiosk.

3. Client-Server Architecture

Client-Server Architecture consists of a centralized server with multiple clients. Its advantages include centralized control, simple deployment, and easy data consistency. Its disadvantages include a single point of failure, scalability bottlenecks, and limited fault tolerance.

Suitability for Coffee Kiosk:

Possible, but the scenario focuses on organizing the internal components of the kiosk, making Layered Architecture more appropriate.

Architecture	Advantages	Disadvantages	Suitability
Layered	Separation of concerns, easy to understand	Performance overhead, tight coupling	Highly suitable
Microservices	Independent scaling, fault isolation	Operational complexity, network latency	Less suitable
Client-Server	Centralized control, simple deployment	Single point of failure, scalability bottleneck	Moderately suitable

Selected Architecture: Layered Architecture

Layered Architecture is selected for the Self-Service Coffee Kiosk because it provides clear separation between presentation, business, and data responsibilities. It is also simpler to manage for this scenario than a Microservices architecture, which introduces additional operational complexity.

COMPONENT TABLE

No.	Component	Responsibility	Layer
1	User Interface Component	Handles touchscreen interaction, coffee selection and size selection	Presentation

No.	Component	Responsibility	Layer
2	Order Manager Component	Manages and processes customer orders	Business
3	Payment Service Component	Processes credit-card payments	Business
4	Receipt Printer Component	Generates and prints order receipts	Business
5	Database Component	Stores menu and pricing information	Data